

MAT 110

Lecture 05

n^{th} derivatives, Leibnitz formula

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Contents

1	Some Useful nth Derivative	1
2	Leibnitz Theorem	2
3	Applications of Leibnitz Theorem	3
4	Recurrence Relations for nth Derivatives	5
5	Practice: Successive Differentiation and Leibniz's Theorem	8
5.1	Successive Differentiation	8
5.2	Leibnitz's Theorem	8

1 Some Useful n th Derivative

Example 01: n th derivative of e^{ax}

Find the n th derivative of

$$y = e^{ax}.$$

Answer.

Let

$$y = e^{ax}.$$

First derivative:

$$y_1 = \frac{dy}{dx} = ae^{ax}.$$

Second derivative:

$$y_2 = \frac{d}{dx}(y_1) = \frac{d}{dx}(ae^{ax}) = a \cdot ae^{ax} = a^2e^{ax}.$$

Third derivative:

$$y_3 = \frac{d}{dx}(y_2) = \frac{d}{dx}(a^2e^{ax}) = a^2 \cdot ae^{ax} = a^3e^{ax}.$$

Continuing this pattern,

$$y_n = \frac{d^n}{dx^n}(e^{ax}) = a^n e^{ax}.$$

Example 02: n th derivative of $\sin(ax + b)$

Find the n th derivative of

$$y = \sin(ax + b).$$

Answer.

Given

$$y = \sin(ax + b).$$

First derivative:

$$y_1 = a \cos(ax + b).$$

Since

$$\cos \theta = \sin \left(\theta + \frac{\pi}{2} \right),$$

we may write

$$y_1 = a \sin \left(ax + b + \frac{\pi}{2} \right).$$

Second derivative:

$$y_2 = a^2 \sin \left(ax + b + \frac{2\pi}{2} \right).$$

Third derivative:

$$y_3 = a^3 \sin \left(ax + b + \frac{3\pi}{2} \right).$$

Therefore, continuing the pattern,

$$y_n = \frac{d^n}{dx^n} \sin(ax + b) = a^n \sin \left(ax + b + \frac{n\pi}{2} \right).$$

n th derivative of $(ax + b)^m$

If

$$y = (ax + b)^m,$$

then the n th derivative is

$$y_n = \frac{d^n}{dx^n} (ax + b)^m.$$

For positive integer m ,

$$y_n = \frac{m!}{(m-n)!} a^n (ax + b)^{m-n}, \quad 0 \leq n \leq m.$$

If $n > m$, then

$$y_n = 0.$$

In particular, if $m = -1$, then

$$\frac{d^n}{dx^n} (ax + b)^{-1} = \frac{(-1)^n n! a^n}{(ax + b)^{n+1}}.$$

2 Leibnitz Theorem

Leibnitz theorem

Leibnitz theorem is a generalization of the product rule. If $u = u(x)$ and $v = v(x)$ are differentiable n times, then

$$\frac{d^n}{dx^n} (uv) = \sum_{k=0}^n \binom{n}{k} \frac{d^k u}{dx^k} \frac{d^{n-k} v}{dx^{n-k}}.$$

Equivalently, if

$$u_r = \frac{d^r u}{dx^r}, \quad v_r = \frac{d^r v}{dx^r},$$

then

$$(uv)_n = uv_n + \binom{n}{1} u_1 v_{n-1} + \binom{n}{2} u_2 v_{n-2} + \binom{n}{3} u_3 v_{n-3} + \cdots + u_n v.$$

Here,

$$\binom{n}{1} = n, \quad \binom{n}{2} = \frac{n(n-1)}{2!}, \quad \binom{n}{3} = \frac{n(n-1)(n-2)}{3!},$$

and in general,

$$\binom{n}{r} = \frac{n(n-1)(n-2) \cdots (n-r+1)}{r!}.$$

Thus,

$$(uv)_n = uv_n + nu_1 v_{n-1} + \frac{n(n-1)}{2!} u_2 v_{n-2} + \frac{n(n-1)(n-2)}{3!} u_3 v_{n-3} + \cdots + u_n v.$$

Special cases

For $n = 1$, Leibnitz theorem gives the ordinary product rule:

$$(uv)' = uv' + u'v.$$

For $n = 2$,

$$(uv)'' = uv'' + 2u'v' + u''v.$$

3 Applications of Leibnitz Theorem

Example 03: Using Leibnitz theorem

Evaluate

$$\frac{d^n}{dx^n} (x^2 e^{ax})$$

using Leibnitz theorem.

Answer.

Let

$$u = x^2, \quad v = e^{ax}.$$

Then

$$u = x^2, \quad u_1 = 2x, \quad u_2 = 2, \quad u_3 = 0,$$

and all higher derivatives of u are zero. Also,

$$v = e^{ax}, \quad v_1 = ae^{ax}, \quad v_2 = a^2 e^{ax}, \quad \dots, \quad v_n = a^n e^{ax}.$$

By Leibnitz theorem,

$$(uv)_n = uv_n + \binom{n}{1} u_1 v_{n-1} + \binom{n}{2} u_2 v_{n-2} + \binom{n}{3} u_3 v_{n-3} + \dots + u_n v.$$

Since

$$u_3 = u_4 = \dots = 0,$$

only the first three terms remain. Therefore,

$$\frac{d^n}{dx^n} (x^2 e^{ax}) = uv_n + \binom{n}{1} u_1 v_{n-1} + \binom{n}{2} u_2 v_{n-2}.$$

Now substituting the values,

$$\frac{d^n}{dx^n} (x^2 e^{ax}) = x^2 (a^n e^{ax}) + \binom{n}{1} (2x) (a^{n-1} e^{ax}) + \binom{n}{2} (2) (a^{n-2} e^{ax}).$$

Since

$$\binom{n}{1} = n, \quad \binom{n}{2} = \frac{n(n-1)}{2},$$

we get

$$\frac{d^n}{dx^n} (x^2 e^{ax}) = a^n x^2 e^{ax} + 2na^{n-1} x e^{ax} + n(n-1)a^{n-2} e^{ax}.$$

Hence,

$$\boxed{\frac{d^n}{dx^n} (x^2 e^{ax}) = e^{ax} (a^n x^2 + 2na^{n-1} x + n(n-1)a^{n-2})}.$$

Example 04: Proving a recurrence relation

If

$$y = \log \left(x + \sqrt{1 + x^2} \right),$$

prove that

$$(1 + x^2)y_{n+2} + (2n + 1)xy_{n+1} + n^2y_n = 0.$$

Answer.

Given

$$y = \log \left(x + \sqrt{1 + x^2} \right).$$

First compute y_1 :

$$y_1 = \frac{d}{dx} \log \left(x + \sqrt{1 + x^2} \right) = \frac{1}{x + \sqrt{1 + x^2}} \left(1 + \frac{x}{\sqrt{1 + x^2}} \right).$$

Now,

$$1 + \frac{x}{\sqrt{1 + x^2}} = \frac{\sqrt{1 + x^2} + x}{\sqrt{1 + x^2}}.$$

Therefore,

$$y_1 = \frac{1}{x + \sqrt{1 + x^2}} \cdot \frac{x + \sqrt{1 + x^2}}{\sqrt{1 + x^2}} = \frac{1}{\sqrt{1 + x^2}}.$$

So,

$$\sqrt{1 + x^2} y_1 = 1.$$

Squaring both sides,

$$(1 + x^2)y_1^2 = 1.$$

Differentiating both sides with respect to x ,

$$\frac{d}{dx} \{ (1 + x^2)y_1^2 \} = 0.$$

Using the product rule and chain rule,

$$2xy_1^2 + (1 + x^2)2y_1y_2 = 0.$$

Dividing by $2y_1$, we get

$$xy_1 + (1 + x^2)y_2 = 0.$$

Therefore,

$$(1 + x^2)y_2 + xy_1 = 0. \tag{1}$$

Now differentiate equation (1) n times with respect to x :

$$\frac{d^n}{dx^n} \{ (1 + x^2)y_2 \} + \frac{d^n}{dx^n} \{ xy_1 \} = 0.$$

By Leibnitz theorem,

$$\frac{d^n}{dx^n} \{ (1 + x^2)y_2 \} = (1 + x^2)y_{n+2} + \binom{n}{1}(2x)y_{n+1} + \binom{n}{2}(2)y_n.$$

Thus,

$$\frac{d^n}{dx^n} \{ (1 + x^2)y_2 \} = (1 + x^2)y_{n+2} + 2nxy_{n+1} + n(n - 1)y_n. \tag{2}$$

Again, by Leibnitz theorem,

$$\frac{d^n}{dx^n} \{xy_1\} = xy_{n+1} + ny_n. \quad (3)$$

Substituting (2) and (3), we get

$$(1+x^2)y_{n+2} + 2nxy_{n+1} + n(n-1)y_n + xy_{n+1} + ny_n = 0.$$

Combining like terms,

$$(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + \{n(n-1) + n\}y_n = 0.$$

Since

$$n(n-1) + n = n^2,$$

we obtain

$$\boxed{(1+x^2)y_{n+2} + (2n+1)xy_{n+1} + n^2y_n = 0}.$$

Hence proved.

4 Recurrence Relations for n th Derivatives

Notation

If

$$y = y(x),$$

then

$$y_1 = \frac{dy}{dx}, \quad y_2 = \frac{d^2y}{dx^2}, \quad y_n = \frac{d^ny}{dx^n}.$$

That means y_n denotes the n th derivative of y with respect to x .

Method for proving recurrence relations

To prove a recurrence relation involving y_n , we usually follow these steps:

1. Start from the given function y .
2. Find a simple differential equation involving y_1 and y_2 .
3. Differentiate that equation n times.
4. Use Leibnitz theorem to expand the derivatives of products.
5. Simplify to obtain the required recurrence relation.

Example 05: Recurrence relation for $y = \tan^{-1} x$

If

$$y = \tan^{-1} x,$$

then show that

$$\boxed{(1+x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0}.$$

Answer.

Given

$$y = \tan^{-1} x.$$

Then

$$y_1 = \frac{1}{1+x^2}.$$

So,

$$(1+x^2)y_1 = 1.$$

Differentiating both sides with respect to x ,

$$\frac{d}{dx} \{(1+x^2)y_1\} = \frac{d}{dx}(1).$$

Thus,

$$(1+x^2)y_2 + 2xy_1 = 0. \quad (1)$$

Now differentiate equation (1) n times with respect to x :

$$\frac{d^n}{dx^n} \{(1+x^2)y_2 + 2xy_1\} = 0.$$

Therefore,

$$\frac{d^n}{dx^n} \{(1+x^2)y_2\} + 2\frac{d^n}{dx^n} \{xy_1\} = 0. \quad (2)$$

Using Leibnitz theorem,

$$\frac{d^n}{dx^n} \{(1+x^2)y_2\} = (1+x^2)y_{n+2} + \binom{n}{1}(2x)y_{n+1} + \binom{n}{2}(2)y_n.$$

So,

$$\frac{d^n}{dx^n} \{(1+x^2)y_2\} = (1+x^2)y_{n+2} + 2nxy_{n+1} + n(n-1)y_n. \quad (3)$$

Again,

$$\frac{d^n}{dx^n} \{xy_1\} = xy_{n+1} + ny_n. \quad (4)$$

Substituting (3) and (4) into (2), we get

$$(1+x^2)y_{n+2} + 2nxy_{n+1} + n(n-1)y_n + 2(xy_{n+1} + ny_n) = 0.$$

Therefore,

$$(1+x^2)y_{n+2} + (2n+2)xy_{n+1} + \{n(n-1) + 2n\}y_n = 0.$$

Since

$$2n+2 = 2(n+1), \quad n(n-1) + 2n = n(n+1),$$

we obtain

$$\boxed{(1+x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0}.$$

Hence proved.

Example 06: Recurrence relation for $y = (\cos^{-1} x)^2$

If

$$y = (\cos^{-1} x)^2,$$

then show that

$$\boxed{(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0}.$$

Answer.

Given

$$y = (\cos^{-1} x)^2.$$

Differentiating with respect to x ,

$$y_1 = 2 \cos^{-1} x \cdot \frac{-1}{\sqrt{1-x^2}}.$$

So,

$$y_1 = -\frac{2 \cos^{-1} x}{\sqrt{1-x^2}}.$$

Hence,

$$\sqrt{1-x^2} y_1 = -2 \cos^{-1} x.$$

Squaring both sides,

$$(1-x^2)y_1^2 = 4(\cos^{-1} x)^2.$$

Since

$$y = (\cos^{-1} x)^2,$$

we have

$$(1-x^2)y_1^2 = 4y.$$

Differentiating both sides with respect to x ,

$$\frac{d}{dx} \{(1-x^2)y_1^2\} = \frac{d}{dx}(4y).$$

Using product rule and chain rule,

$$(-2x)y_1^2 + (1-x^2)2y_1y_2 = 4y_1.$$

Dividing by $2y_1$,

$$-xy_1 + (1-x^2)y_2 = 2.$$

Therefore,

$$(1-x^2)y_2 - xy_1 - 2 = 0. \tag{1}$$

Now differentiate equation (1) n times with respect to x , where $n \geq 1$:

$$\frac{d^n}{dx^n} \{(1-x^2)y_2\} - \frac{d^n}{dx^n} \{xy_1\} - \frac{d^n}{dx^n}(2) = 0.$$

Since

$$\frac{d^n}{dx^n}(2) = 0 \quad (n \geq 1),$$

we get

$$\frac{d^n}{dx^n} \{(1-x^2)y_2\} - \frac{d^n}{dx^n} \{xy_1\} = 0. \tag{2}$$

Using Leibnitz theorem,

$$\frac{d^n}{dx^n} \{(1-x^2)y_2\} = (1-x^2)y_{n+2} + \binom{n}{1}(-2x)y_{n+1} + \binom{n}{2}(-2)y_n.$$

So,

$$\frac{d^n}{dx^n} \{(1-x^2)y_2\} = (1-x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n. \tag{3}$$

Also,

$$\frac{d^n}{dx^n} \{xy_1\} = xy_{n+1} + ny_n. \tag{4}$$

Substituting (3) and (4) into (2), we get

$$(1 - x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n - (xy_{n+1} + ny_n) = 0.$$

Thus,

$$(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} - \{n(n-1) + n\}y_n = 0.$$

Since

$$n(n-1) + n = n^2,$$

we obtain

$$(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0.$$

Hence proved.

5 Practice: Successive Differentiation and Leibniz's Theorem

5.1 Successive Differentiation

Practice A: Find the n th derivative

Find the n th derivative of the following functions:

1. $y = x^n$.
2. $y = (ax + b)^n$.
3. $y = \ln(ax + b)$.
4. $y = \frac{1}{x + a}$.
5. $y = e^{ax}$.
6. $y = \sin(ax + b)$.
7. $y = \cos(ax + b)$.

5.2 Leibniz's Theorem

Practice problems on Leibniz's theorem

Use Leibniz's theorem and successive differentiation to prove the following results.

1. If $y = \tan^{-1} x$, then show that $(1 + x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$.
2. If $y = \cot^{-1} x$, then show that $(1 + x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$.
3. If $y\sqrt{1 - x^2} = \sin^{-1} x$, then show that $(1 - x^2)y_{n+1} - (2n+1)xy_n - n^2y_{n-1} = 0$.
4. If $y = e^{\tan^{-1} x}$, then show that $(1 + x^2)y_{n+2} + \{2nx + 2x - 1\}y_{n+1} + n(n+1)y_n = 0$.
5. If $y = e^{m \sin^{-1} x}$, then show that $(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 + m^2)y_n = 0$.
6. If $y = (\sin^{-1} x)^2$, then show that $(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$.
7. If $\ln y = a \sin^{-1} x$, then show that $(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 + a^2)y_n = 0$.
8. If $y = e^{m \cos^{-1} x}$, then show that $(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 + m^2)y_n = 0$.
9. If $\ln y = \tan^{-1} x$, then show that $(1 + x^2)y_{n+2} + \{2nx + 2x - 1\}y_{n+1} + n(n+1)y_n = 0$.
10. If $y = (\cos^{-1} x)^2$, then show that $(1 - x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$.

11. If $\ln y = m \cos^{-1} x$, then show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$.
12. If $x = \tan(\ln y)$, then show that $(1 + x^2)y_{n+2} + \{2nx + 2x - 1\}y_{n+1} + n(n + 1)y_n = 0$.